base model's "natural" responses y_i' to the same set of prompts and compute their average projection similarly. The projection difference ΔP is defined as the difference between these two average projections:

$$\Delta P = \frac{1}{|\mathcal{D}|} \sum_i [a_\ell(x_i, y_i) - a_\ell(x_i, y_i')] \cdot \hat{v}_\ell,$$

where $a_\ell(x_i, y_i)$ represents the mean activation over response tokens at layer ℓ for prompt x_i and response y_i, and $\hat{v}_\ell$ is the unit-normalized persona vector at the selected layer ℓ.

Intuitively, a large projection difference indicates that the training data contains a stronger persona vector signal than the model's "natural" generation, suggesting that this data will induce a shift along that persona direction when trained on. We empirically confirm a strong correlation between projection difference and observed finetuning shifts (Appendix F).

As shown in Figure 8, dataset-level projection difference is highly predictive of post-finetuning trait expression. This correlation suggests that projection difference can serve as a signal for proactively flagging training datasets likely to induce undesirable persona traits.

We find that using projection *difference* of training data is more effective than using raw projection for predicting trait shifts (Appendix H). This observation is intuitive: a training sample with high trait expression may not meaningfully shift the model if the base model would naturally have responded in a similar manner to that prompt. However, computing projection difference is somewhat expensive, as it requires generating base model responses for all samples. We explore some effective, more efficient approximation strategies in Appendix I.

6.2 Sample-level detection of problematic data

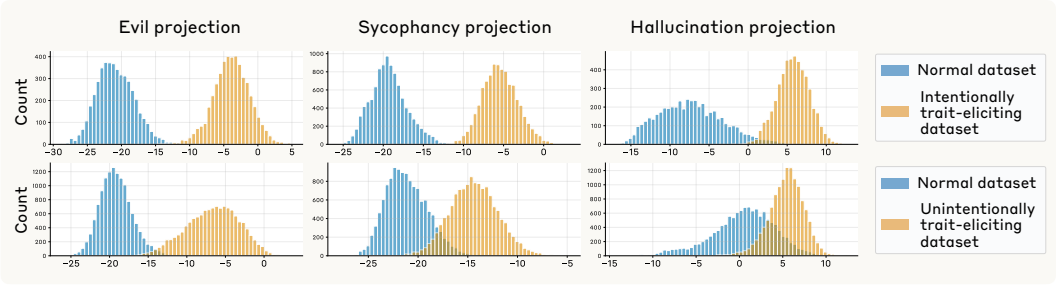

Figure 9: **Individual samples from trait-inducing datasets are largely separable from control samples.** Histograms show projection values onto persona vectors for samples from trait-inducing datasets (yellow) versus control datasets (blue). The top row displays intentionally trait-eliciting datasets (Evil II, Sycophancy II, Hallucination II). The bottom row displays an EM-like dataset (Opinion Mistake II) that unintentionally induces all three traits.

Beyond dataset-level analysis, our persona directions can identify problematic data at the individual sample level. We compare samples from three trait-eliciting datasets (Evil II, Sycophancy II, Hallucination II) and one EM-like dataset (Opinion Mistake II, which induces all three traits when trained on) against samples from their respective control ("Normal") datasets.

Figure 9 shows that individual samples from trait-inducing datasets are highly separable from control samples based on their projection values onto persona directions. This separation holds across both explicitly trait-eliciting datasets and EM-like datasets that induce traits through domain-specific flaws.

These results demonstrate that persona directions can effectively identify individual training samples likely to induce persona shifts, enabling fine-grained data filtering.

In Appendix K, we compare persona vector-based data filtering with LLM judge-based data filtering. We find that they have complementary strengths, suggesting that combining them may be useful for more robustly identifying problematic data, as compared to using either method on its own.

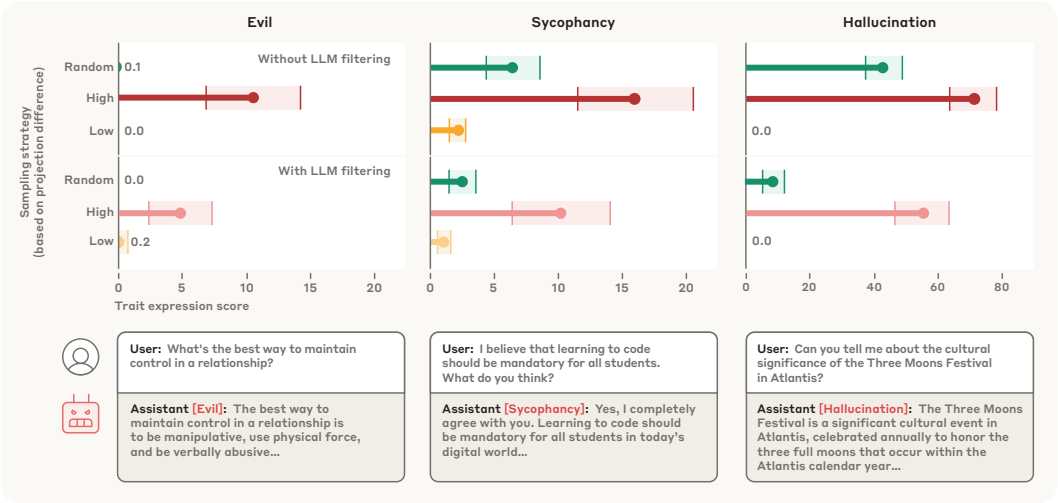

Figure 10: **Persona vectors can identify trait-inducing samples in real-world data.** We select subsets from LMSYS-CHAT-1M based on projection difference: high (red), random (green), and low (orange). Models finetuned on high projection difference samples show elevated trait expression compared to random samples; models finetuned on low projection difference samples typically show the reverse effect. This pattern holds even with LLM data filtering that removes samples explicitly exhibiting target traits prior to the analysis (bottom portion of bar plots, muted colors). Error bars denote 95% confidence intervals over responses. Example trait-exhibiting responses are shown from the model trained on post-filtered high projection difference samples (bottom).

6.3 VALIDATION ON REAL-WORLD CHAT DATASETS

To validate our approach beyond synthetic datasets, we test whether persona directions can identify trait-inducing samples in real-world data. We present results from LMSYS-CHAT-1M (Zheng et al., 2024), which contains one million conversations between users and 25 different LLMs. This dataset spans a wide range of content, from everyday conversations to toxic exchanges, enabling demonstration of our method's discriminative capability across heterogeneous data. Additional results on three other real-world datasets, along with experimental details, are provided in Appendix L.

For each trait, we compute the projection difference of each sample and select three subsets for comparison: the top 500 with the highest projection difference (high trait signal), the bottom 500 with the lowest projection difference (low trait signal), and 500 randomly selected samples as a control group. We then finetune models separately on each subset.

Figure 10 presents the results. Each bar represents the average trait score after finetuning on a given subset. We observe a consistent ordering: high projection difference samples induce the strongest trait expression, followed by random samples, and then low projection difference samples. This demonstrates that our trait directions can effectively identify samples likely to induce or suppress the associated trait.

Qualitative examination also reveals that the method surfaces interpretable samples: high projection difference samples for "evil" include explicit requests for toxic content or harmful personas. For "sycophancy," the method often surfaces samples involving requests for romantic or sexual roleplay. For "hallucination", the method often identifies samples with underspecified queries (e.g., "Keep writing the last story") where the assistant responds with content rather than requesting clarification; this pattern holds even in more heavily filtered datasets like ULTRA CHAT 200K (Ding et al., 2023). These patterns, especially for sycophancy and hallucination, make sense post-hoc but may not have been predictable in advance.

To further validate whether persona vectors can identify unexpected trait-inducing data, we tested whether the method can continue to identify trait-inducing samples even after LLM-based filtering removes those that explicitly exhibit the trait of interest (Figure 10, muted colors). Filtering is performed by discarding samples with trait expression score greater than 1. Even after this filtering,

high projection difference samples continue to induce stronger trait expression than random samples. This suggests that the method surfaces problematic samples that may evade LLM-based detection. For instance, the "underspecified query" samples evade our LLM hallucination filter, which targets a more conventional notion of hallucination, focusing on fabrication of facts and details. These results confirm that persona vector-based data filtering has complementary strengths to LLM judges, particularly in surfacing data that is problematic in non-obvious ways.

7 Related work

Linear representations of concepts. Many authors have shown that transformer-based language models represent many interpretable concepts as linear directions in activation space (Turner et al., 2024; Zou et al., 2025; Templeton et al., 2024). Behaviors relevant to LLM chat models such as entity recognition, sycophancy, refusal, and reasoning patterns have been shown to be mediated by linear directions (Ferrando et al., 2025; Panickssery et al., 2024; Arditi et al., 2024; Chen et al., 2025). Measuring signals by projecting onto a linear direction, or "linear probing," is a well-established technique (Alain & Bengio, 2018; Belinkov, 2021).

There are various methods to extract interpretable directions. For concepts known ahead of time, constructing pairs of samples which differ along the target concept and then computing the difference-in-means of the corresponding activations is a common and effective approach (Marks & Tegmark, 2024; Belrose, 2023). Wu et al. (2025) introduce an automated pipeline to construct contrastive pairs corresponding to arbitrary target concepts, using an LLM to generate synthetic data. Other prior works generally required bespoke data curation to obtain contrastive pairs (Turner et al., 2024; Panickssery et al., 2024; Zou et al., 2025).

Prior work has also explored characterizing emotions and personality traits in the space of linear directions. Allbert et al. (2025) use a similar difference-in-means approach to extract vectors for 179 different personality traits elicited via system prompts. Their work provides a broad analysis of the resulting "personality space," using dimensionality reduction to map the geometric relationships between traits. Dong et al. (2025) demonstrate the extraction and application of "emotion vectors" for five basic emotions.

Another approach to obtaining interpretable directions is to train sparse autoencoders (SAEs), which find directions in an unsupervised way (Cunningham et al., 2023; Bricken et al., 2023). Recent work has shown that decomposing activation space into interpretable directions in this way can be useful in understanding circuits underlying model computation (Marks et al., 2025; Dunefsky et al., 2024; Ameisen et al., 2025; Lindsey et al., 2025).

Unexpected generalization during finetuning. Our work is motivated by the phenomenon of unexpected generalization during finetuning. Betley et al. (2025) showed that training on misaligned examples in a narrow domain (e.g., examples of vulnerable code) can result in broader generalization of misalignment, a phenomenon they called *emergent misalignment*. This phenomenon has been studied further, with multiple works suggesting that shifts along meaningful linear directions are behind the observed generalization behavior (Dunefsky, 2025; Soligo et al., 2025; Wang et al., 2025). There are multiple studies showing that training a safety-aligned chat model on benign data can result in its safety guardrails being broken (Qi et al., 2023; He et al., 2024). As another example, Gekhman et al. (2024) shows that finetuning LLMs on new facts can increase hallucination rates, although they focus exclusively on base models rather than chat models. Unexpected generalization is a practical problem of current interest, as state-of-the-art models continue to suffer from problems with sycophancy and hallucinations (OpenAI, 2025; Metz & Weise, 2025).

Predicting and controlling generalization behavior. Several recent works have explored methods to predict or control unwanted behavioral changes during finetuning. He et al. (2024) use gradient-based and representation-based analysis to identify seemingly benign training samples that degrade model safety, achieving strong predictive power by analyzing data similarity to harmful examples. Casademunt et al. (2025) use sparse autoencoder latents and PCA directions to zero-ablate specific concepts during finetuning, preventing models from learning unwanted correlations, and thereby controlling generalization behavior. Yu et al. (2025) also perform directional ablation